Humanities & Social Sciences Communications

ARTICLE

https://doi.org/10.1057/s41599-023-02098-8 OPEN

Understanding customer experience with Vietnamese hotels by analyzing online reviews

Ha Thi Thu Nguyen [1]✉ & Trung Xuan Nguyen [2]✉

On the post-pandemic recovery green economy track, especially in the hospitality and tourism industry, businesses must accelerate digital transformation and enter the race to increase customer experience. Improving customer experience is critical, and understanding customers' emotions and needs after using hotel services is crucial. This is why businesses increasingly rely on online customer reviews on booking sites to gain insight into what their customers seek. By understanding their needs and experiences, hotel managers can provide services that meet customer expectations and improve overall service quality. Although online review analysis is meaningful for hotel managers because of its practicality in business operations, a specific model, method, or tool is needed to explore customer emotions. This study proposes a method to analyze customers' online reviews of Vietnamese hotel services. With a Data set of 20,551 reviews collected from TripAdvisor, the study examines customers' perceptions of Vietnamese hotel services overall and aspects of hotel services by combining language rules of natural language and inferential statistics. The key findings indicate that customers are most satisfied with the "place" aspect of hotels, with a satisfaction rate of 78%, while the "room" service aspect has the lowest satisfaction rate at 61.3%. These results have recommended that hotel managers in Vietnam prioritize understanding customers' sentiments and opinions to improve service quality in aspects where satisfaction rates are lower.

[1] Department of Greenwich, FPT University, Hanoi, Vietnam. [2] Faculty of Management Science, Vietnam Graduate Academy of Social Sciences, Hanoi, Vietnam. ✉email: hantt194@fe.edu.vn; trungnguyenxuan@gmail.com

Introduction

Customer experience has been a concern for businesses and researchers for over 20 years (Akhtar et al., 2017; Rahimian et al., 2021). According to studies, consumer experience is defined using a psychological, anthropological, ethnographic, marketing, or economic point of view (Breda et al., 2020; Luturlean & Anggadwita, 2016; Paulose & Shakeel, 2022). While customer experience is personal and multi-dimensional, managing it is a business issue (Luturlean & Anggadwita, 2016; Mohamed, 2021). Effective customer experience management involves understanding customer emotions and implementing strategies to enhance their experience (Luturlean & Anggadwita, 2016). This helps businesses gain insight into customer psychology, emotions, and opinions, identify areas for improvement, and develop plans to enhance product quality and customer care (Lee et al., 2019; Paulose & Shakeel, 2022). Researchers have found that offering cheaper prices or better products and services is no longer an effective competitive strategy. Instead, businesses must focus on creating a unique "customer experience" (Mohamed, 2021). In Vietnam, measuring customer experience is one of the six main pillars used to evaluate business digital transformation, as outlined by the Ministry of Information and Communications. As a result, companies are now prioritizing customer experience management.

The growth of the Internet has positively impacted the hotel industry, as it has led to the development of new services and business models to meet customers' changing needs. Providing an excellent customer experience is crucial for hotels to build a loyal customer base, enhance their reputation, and increase revenue (Paulose & Shakeel, 2022; Sampetua Hariandja, Vincent (2022)). As a result, many scholars and hotel managers are interested in researching customer experience in this field (Blomberg-Nygard, A., & Anderson, C. K., 2016). Just like in other industries, customer experience research in the hospitality sector examines the impact of customer experience on emotions, psychology, perspectives, and relationship-building (Sampetua Hariandja, Vincent (2022)).

Under the influence of e-commerce and sharing economy, online booking sites are also developing (Alrawadieh & Law, 2019; Chalupa & Petricek, 2022; Dong et al., 2014). Recent customer experience studies are turning to data analysis of online customer reviews (Li et al., 2019; Li et al., 2020). The trend of studies from 2014 has diverted research focuses on analyzing customer-generated big data on the Internet because of its high availability, popularity, and large amount of data (Hutto, Gilbert (2014); Zhao et al., 2019; Alaei et al., 2019; Barnes et al., 2020; Chalupa & Petricek, 2022). Data analytics allows for the development of new techniques to extract valuable insights from large volumes of data (Barnes et al., 2020). Data analytics captures environmental changes in the tourism and hospitality industries and prepares long-term plans and strategies (Chen et al., 2019). Unlike traditional approaches, data analysis supports data-driven decision-making (Godnov & Redek, 2018; Gómez-Suárez & Veloso, 2022). Analyzing customer reviews in text form also can help identify keywords representing the experience and underlying patterns (Hu et al., 2019; Li et al., 2019; Hutto, Gilbert (2014)). Text analysis of hotel experiences customers create reveals their true emotions (Hu et al., 2019; Li et al., 2019; Hutto, Gilbert (2014)). Some characteristics of the data analytics approach are:

- Customer-centricity brings high efficiency when managing relationships and taking care of customers.
- Analyze customer psychology with many objective dimensions, not limited based on pre-designed questions.
- Data is available online, saving costs for new data collection plans.
- The model is challenging in initial construction but to can be applied to almost all hotels when completed.
- Data is updated quickly to support managers with more information when making administrative decisions.

Vietnam's economy is bouncing back after experiencing the effects of Covid for two years. The country boasts a coastline exceeding 3000 km and is home to nearly a dozen UNESCO-recognized landscapes, giving it immense tourism potential (Thu et al., 2020). The tourism industry has witnessed a significant resurgence since the end of 2022, with a considerable increase in visitors. The hotel industry is vital in developing Vietnam's tourism and accounts for over 5% of the national GDP (Thu et al., 2020). As a result, the Vietnamese Government is keen on improving service quality to attract more customers. Therefore, finding solutions for measuring customer experiences and satisfaction with hotel services and enhancing quality to meet customer needs is an important issue (Thu et al., 2020).

Previous studies have discussed methods for measuring customer experience, but no established formulas, algorithms, or processes have clearly described the measuring method (Thu et al., 2020). In Vietnam, hotel managers still need to utilize data analysis methods to measure customer experience with hotel services. Instead, they often rely on interviews or questionnaires for surveys (Vo et al., 2022). However, this approach is limited by sample size and is only appropriate for individual hotels, making it difficult to scale for larger establishments. Therefore, this study aims to answer the research question: "Can standard formulas combined with lexical rules be developed to measure customer satisfaction based on data analysis?" This study will present a series of definitions, formulas, and algorithms to describe the measuring process clearly.

This study gathered data from 12 hotels in six major cities in Vietnam, including Hanoi, Ho Chi Minh, Danang, Hue, Quy Nhon, and Nha Trang. After pre-processing, the data consisted of 20,551 reviews. We utilized the Python language and Vader library for natural language analysis, along with lexical rules, to complete the assigned tasks:

- Forming a series of definitions and measurement formulas that have not been explicitly mentioned in previous studies
- Measuring overall customer satisfaction with hotel services in Vietnam
- Statistics inferential method combined with lexical rules to deeply discover the relationship between aspect and polarity words. Measuring customer satisfaction with service aspects.

The paper is structured as follows: Part 2 presents research on customer experience with hotel services based on data analysis. Part 3 introduces the data analysis approach using the Python library. Section 4 explains the research method, section 5 presents the measurement results, and the paper concludes with a final section.

Literature review

Natural language processing and sentiment analysis. Natural Language Processing (NLP) is a branch of artificial intelligence that helps computers understand, interpret, and manipulate human language. NLP crosses multiple disciplines, including computer science and computer linguistics, to fill the gap between human communication and computer knowledge (Hu et al., 2019; Chowdhary, 2020; Dominy et al., 2022; Liu et al., 2017).

Natural language processing can solve many real-life problems, such as text summarization, automatic translation, sentiment analysis, etc. Understanding customer opinion using NLP is being used by companies and organizations (Hutto, Gilbert (2014); Akhtar et al., 2017; Alaei et al., 2019; Breda et al., 2020). The companies focus on research to optimize personalized services (Rahimian et al., 2021). Analyzing customer sentiment will help

managers to make better decisions (Rahimian et al., 2021; Godnov & Redek, 2018; Luturlean & Anggadwita, 2016). It includes measuring customer satisfaction with the brand, products, and services (Thu et al., 2020; Sampetua Hariandja, Vincent (2022); Gómez-Suárez & Veloso, 2022). It can explore opinions, feelings, subjective conceptions, evaluations, attitudes, appraisals, emotions, and so forth expressed in the text (Gómez-Suárez & Veloso, 2022). Data for mining through comments, blogs, discussions, news, comments, feedback, or other materials (Gómez-Suárez & Veloso, 2022; Hutto, Gilbert (2014); Dominy et al., 2022). Opinion mining is based on computational languages, information retrieval, text data mining, natural language processing, machine learning, statistics, and predictive analytics (Li et al., 2019; Alaei et al., 2019; Akhtar et al., 2017).

Extracting hotel service aspects from online reviews. Customers often leave their reviews on e-commerce and booking websites to express their opinions on products and services. These reviews are typically unstructured data; researchers have analyzed them to understand customer sentiments (Chen et al., C019; Li et al., 2019; Lee et al., 2019). Hotel managers can also use these reviews to gain insight into customer psychology through positive and negative feedback (Chalupa & Petricek, 2022). This approach transforms qualitative data into quantitative measures (Chowdhary, 2020; Barnes et al., 2020; Thu et al., 2020), measuring customer satisfaction with service quality. Additionally, specific aspects of hotel services that make customers happy can be identified (Akhtar et al., 2017; Alrawadieh & Law, 2019; Dong et al., 2014; Li et al., 2020; Liu et al., 2017; Paulose & Shakeel, 2022). For instance, customers may be satisfied with the hotel staff, location, and rooms.

Example 2. Suppose that we have a customer review about the hotel as follows:

"The service was great, but the food wasn't that good."

The following set of words assembles the above sentence:

["The", "service", "was", "great,", "but", "the", "food", "wasn't", "that", "good."]

In this review sentence, two aspects of the hotel mentioned by customers can be identified: "service" and "food". Both of these aspects are nouns.

The customer's opinion is expressed as satisfaction or dissatisfaction with a particular aspect. It has positive or negative connotations. To express positivity or negativity (or call polarity) in this sentence are usually adjectives and accompanying adverbs. For example, "great" and "good" are two adjectives, and "not" is an adverb.

In this sentence, the customer's opinion expresses satisfaction with the service and dissatisfaction with the hotel's food.

Therefore, extracting attributes (also called aspects) from customer reviews is to find keywords representing hotel aspects for measuring quality. These aspects are nouns. Hotel services are often mentioned in customer reviews; their frequency will be high if many customers mention them. It also expresses the customer's interest when it is mentioned many times.

Call R is the set of customer reviews $R = \{r_1, r_{2,\ldots,}r_n\}$

then the attribute set of the hotel is the set of keywords $W = \{w_1, w_{2,\ldots,}w_n\}$ that are extracted from R with w_i is a noun and ensure that $f(w_i) > k$, in which $f(w_i)$ is the frequency of keywords and k is a bias of specific value, and it is used to select important keywords.

The set of attributes of this hotel is combined with several other parameters to become a set of factors affecting customer satisfaction. (Thu et al., 2020).

Measuring Customers' experience with hotels using NLP. Today's online booking service platforms are also a vast source of big data that provides the tourism industry with a free accessible way (Breda et al., 2020). The enormous amount of hotel information data provided to hotel managers runs into millions of records. Another significant unstructured data source is each hotel's customer reviews, similar to a social network. The analysis to understand these data helps both hotel managers and customers (Chen et al., 2019).

In recent years, researchers have explored the use of data mining to understand hotel experiences based on customer-generated content online (Mohamed, 2021). To achieve this, they have used text mining software to extract frequently used words that represent important aspects of the customer experience, such as service quality topics like "restaurant", "breakfast", "food", "pool", "reservation", "buffet", "lounge", "bar", "check-in", "floor", "view", "cleanliness", "bed", "bathroom", and "staff" (Zhao et al., 2019; Hutto, Gilbert (2014)). This approach allows researchers to evaluate guest satisfaction or dissatisfaction with various hotel services. Table 1 presents a list of case studies related to this approach.

These studies examine how analytics on hotel customer experience data, generated by customers themselves on online travel or booking sites, can be used to uncover customer sentiments, emotions, and perspectives (Zhao et al., 2019; Akhtar et al., 2017; Alrawadieh & Law, 2019; Breda et al., 2020; Leal et al., 2019). By leveraging this data, businesses can identify new opportunities, understand customer experiences, improve service quality, and develop better customer care plans. Ultimately, this can attract more tourists to the hotel (Chalupa & Petricek, 2022).

The studies mentioned above all utilize NLP in various ways. Some employ LDA or a similar method to identify and group topics, while others use word frequency statistics (Hu et al., 2019; Leal et al., 2019; Li et al., 2020; Hutto, Gilbert (2014)). However, none of the studies utilize lexical rules or address how to handle the combination of nouns and polarity words (adjectives and adverbs). Additionally, while the studies describe the methods used, they do not mention the development of standard formulas or design measurement methods or ors on practical problem-solving steps. Some recent studies have only surveyed a small number of samples (Godnov & Redek, 2018; Alrawadieh & Law, 2019), rendering their results non-universal. They only measure aspects in satisfied sentences without combining aspect and polarity words (Godnov & Redek; Li et al., 2020; Alrawadieh & Law, 2019). Some methods utilize independent variables to describe the relationship with customer satisfaction (Zhao et al., 2019). Furthermore, they found that long sentences or other variables impacted overall ratings (Zhao et al., 2019).

Natural language processing with Python

Python language and NLP libraries. Python has many applications, including web development, software development, data science, and machine learning (ML). It was developed by Guido Van Rossum in 1989 at Centrum Wiskunde & Informatica. Python is highly efficient and compatible with various platforms.

Python has an extensive collection of libraries that cater to non-programming developers. With over 137,000 libraries available, Python can support various applications in the field of data science and machine learning (Hutto & Gilbert (2014)). For natural language processing, Python offers the NLTK library, which provides function libraries, analysis tools, corpus, wordnet, and more to simplify and expedite the process. NLTK is a powerful and practical toolkit for Natural Language Processing with numerous features to help analyze data: text can be divided into words or sentences, stop words can be filtered, word types can be labeled, and word frequency can be calculated (Chowdhary, 2020).

For example: To remove stop words can use the following library

```
import nltk
from nltk.corpus import stopwords
```

Table 1 Some related studies.

Related study	Data source	Methodology	Findings
Dong et al. (2014)	24,051 online hotel reviews were collected from Daodao.com with 4–5 stars of 100 hotels.	- Use ROST CM6.0 tool for mining Chinese text. - From Chinese text mining to understand how satisfaction of customer. - Grouping synonyms and word frequency statistics	- Discover the most frequently mentioned frequencies in the study: Hotel (25.9%), Location (21.7%), Room (21.4%), Service (14.3%), Dining & Food (8.1%), Value (4.8%), Facility availabilities (3.8%). - Discover combinations of polarities to predict customer satisfaction
Hu et al. (2019)	27,864 hotel reviews on Tripadvisor	Using STM for investigating user dissatisfaction, which overcomes the limitations of studies based on the LDA topic model and word frequency analysis	- 10 negative topics are based on statistical that these topics' appearances in negative reviews are substantially higher than in positive reviews. - For low-level hotels, facilities are the main cause of customer dissatisfaction. High-level hotel prices issue is the main antecedent of customer dissatisfaction. - Explore a list of negative words in customer complaints in separate topics: Facilities (noisy, loud, air, heat, unit, sound, hear), Service (smoke, Priceline, double, nonsmoking, twin), Location (distance, central,within, block, subway)
Godnov & Redek (2018)	1712 online reviews of 28 different hotels in Slovenian on TripAdvisor	- Use a value-based approach to each word in a sentence using AFINN Lexicon - Identify the aspect words in the sentence and use Aylien software and R programming to identify and evaluate the positive, negative, and neutral values in each aspect. - Use LDA (Latent Dirichlet Allocation) to separate words, identify topics, and group topics	- Assess general satisfaction with a specific location (seaside, ski, or spa) - Separating words and calculating the frequency of topic words and polarity words - Separate topic groups by specific location - Measure aspects with positive rating
Alrawadieh & Law (2019)	400 English-language reviews about the hotels in Istanbul, Turkey, that collected from TripAdvisor's	- Explore aspects of guest satisfaction uncaptured by traditional research methods	- Divided into categories: Room (satisfaction: 29%, dissatisfaction: 19.1%), Service quality (satisfaction: 29.1%, dissatisfaction: 28%), Hotel's characteristics (satisfaction: 26%, dissatisfaction: 41.1%), Food and beverage (satisfaction: 15%, dissatisfaction: 11.8%), - The room quality in terms of size, furniture, amenities, location, bathroom, cleanliness, and design was the most mentioned issue by reviewers. - The content analysis of 400 online travel reviews produced 1,664 positive and 236 negative elements, grouped under four main constructs, namely, rooms, service quality, hotel characteristics, and Food and beverage.
Li et al. (2019)	Extracting 412,784 consumer-generated reviews from TripAdvisor across different cities in China	- Software of R was used for data analysis to inspect the asymmetric effects of focal hotel attributes on customer satisfaction and dissatisfaction - Overall rating is calculated by variables: high cleanliness rating, low cleanliness rating, high value rating, low value rating, high room rating, low room rating, high service rating, low service rating, high location rating, low location ratings, city	- Explores the asymmetric effects of hotel attributes on customer satisfaction: Location, Room, Service, Value, and Cleanliness. - Guests' expectations of hotel performance differ with respect to their origins and the star ratings of the hotels being reviewed, thereby moderating the asymmetric impact of hotel attributes on customer satisfaction.
Zhao et al. (2019)	127,629 reviews from tripadvisor.com About 217 hotels in San Francisco	NLTK package in Python programming, TextBlob Python library for sentiment analysis. - Overall customer satisfaction through variables: Subjectivity, Diversity, Readability, Polarity, Length, Involvement, Hotel Ranking	Variables with higher levels of subjectivity and readability and a longer length of textual review lead to lower overall customer satisfaction and a higher level of diversity and sentiment polarity of textual review leads to higher overall customer satisfaction

Sentiment analysis of experienced Customers by VADER library. There are a variety of opinion and sentiment analysis libraries available in Python, one of which is the VADER library. VADER stands for Valence Aware Dictionary and Sentiment Reasoner, and it is an open-source, rule-based opinion analysis library in NLTK. The library measures the semantics in the text and produces a probability value for each sentence, classifying it as positive, negative, or neutral. This tool can analyze text data from social media platforms like Twitter and Facebook. VADER has a high level of accuracy and performs even better than human evaluation on the Tweeter dataset (with a classification accuracy F1 of 0.96 and 0.84, respectively). For the Movie reviews dataset, it has an overall score of 0.7, which is lower than human evaluation but higher than other methods like Hu-L, SentiWordNet (SWN), SenticNet (SCN), Word-Sense Disambiguation (WSD) using WordNet (Hutto & Gilbert (2014)).

Features of VADER:

- No training data is required
- Understand the opinion of text containing emojis, slang, conjunctions, etc.
- Optimized for text on social networks
- Open Source Library

Example 1. Give a sentence
"The movie was awesome!"
VADER can analyze this sentence into Positive: 99% Negative: 1% Neutral: 0%

Table 2 List of aspect group.

Location	Room	Check in	Services	Staff	Meal	Surrounding	Value
Place	Bed	Front - desk	Swimming -pool	housekeeping	Breakfast	Restaurant	Price
Town	Dirty	Check - out	Spa	Staffer	Food	Super -Market	Worth
Area	Toilet	Lounge	Massage	Personnel	Buffet	Area	Cost
Close	View	Hall	Wireless	Waitress	Bar	Central	Amount
Central	Bathroom	Reception	Gym	Cashier	Restaurant	Hospital	Saving
Airport	TV	Greeting	Internet	Receptionist	Tea	River	Discount
Distance	Television	Guest	Parking	Manager	Coffee	Beach	Profits
Village	Fridge		Gymnasiums	Doorman	Luch	Street	Sales
Street	Floor		Wifi	Bookkeeper	Dinner	Landmark	valuation
Province	Boardroom		Supporting		Brunch	Temple	
						Lake	

Methodology

The fundamental concepts

Definition 1. Set of experienced guest

A set of experienced guests are those who have used hotel services in Vietnam and provided feedback on TripAdvisor. Each guest has a review r_i. So, it is expressed by:

$$\mathcal{G} = \{g_1, g_2, \ldots, g_m\} \quad (1)$$

Definition 2. Satisfaction

Satisfaction of the guest g_j is who wrote review r_i and it is measured by sentiment value of review r_i with a combination function λ

$$Sas\left(g_j\right) = \lambda.\left(neg\left(r_i\right) + pos\left(r_i\right) + neu\left(r_i\right)\right) \quad (2)$$

Definition 3. Aspects

During a guest's stay at a hotel, aspects refer to the services provided, including the room, staff, and the hotel's location. Aspects are categorized as nouns.

$$\mathrm{A} = \{a_1, a_2, \ldots, a_m\} \quad (3)$$

With a_k is the aspect of the hotel.

When customers write reviews, they may use different words to describe the same aspect. To extract these aspects, similar words are grouped together based on their meaning. Previous studies, such as the one by Godnov and Redek (Godnov & Redek, 2018) have used a technique called LDA to group words according to their topic. In this paper, we use aspect groups that have already been extracted from research conducted by Thu et al. (Thu et al., 2020). Table 2 shows the list of these aspect groups.

In this group of aspects, for example, Location has a set of similar words: place, town, area, close, central, airport, distance, village, street, and province. Therefore, each aspect has a set of terms with equivalent meanings.

$$a_k = \{a_{k1}, a_{k2}, \ldots, a_{kr}\}$$

With a_k is the aspect k^{th} and a_{ki} is the aspect i^{th} of aspect a_k

The process of measuring customer experience. We analyze customer feedback on TripAdvisor to measure customer experience in Vietnamese hotels. This feedback reflects their sentiments and level of satisfaction. When analyzing customer satisfaction, there are two main measurement tasks to consider as below:

- Overall satisfaction shows that the customer is satisfied with the hotel service.
- Satisfaction by each service (aspect) shows that customers are satisfied with each hotel service at a detailed level (room, staff, meal, location, etc).

The proposed method for measuring customer satisfaction in this study is illustrated in Fig. 1, as described below:

Our process of research consists of three steps that begin with gathering data and end with assessing the outcome:

- Step 1. Data collection: The first step involves collecting customer review data on Vietnam hotels from TripAdvisor using an automatic collection tool. The data is then stored in a .csv file.
- Step 2. Data storage and processing: data processing eliminates patterns that do not conform.
- Step 3. Data analysis:
  - Measure the overall satisfaction of hotel customers using Python.
  - We use Python, a natural language processing library, and a list of aspect groups (see Table 2) to identify the hotel aspects mentioned by customers and analyze their satisfaction and dissatisfaction from reviews.
  - We use algorithmic models and tools to analyze the quantified data and present the method's results.

Measuring overall satisfaction of customers with hotel services. The overall satisfaction rate indicates the percentage of customers satisfied with hotel services in Vietnam. The following formula is used to calculate:

$$Sas_{overall} = \frac{number\ of\ positive\ reviews}{total\ of\ reviews} \times 100\% \quad (4)$$

In which:

- Number of positive reviews: number of positive reviews
- Total of reviews: total of reviews

These are the steps we take to measure:

- Step 1. Use the Vader library to measure customer reviews by calculating positive, negative, and neutral scores. Then aggregate these values by a unique score. In which overall customer satisfaction is measured according to formula (2). In Python, this is represented by the following code:

```
apply(lambda review: analyser.polarity_
scores(review))
```

- Step 2. We select positive reviews and use formula (4) to measure customer satisfaction with Vietnamese hotel services.

Measuring the satisfaction of customers with hotel aspects. To measure customer satisfaction with different aspects of a hotel, two sub-tasks can be performed:

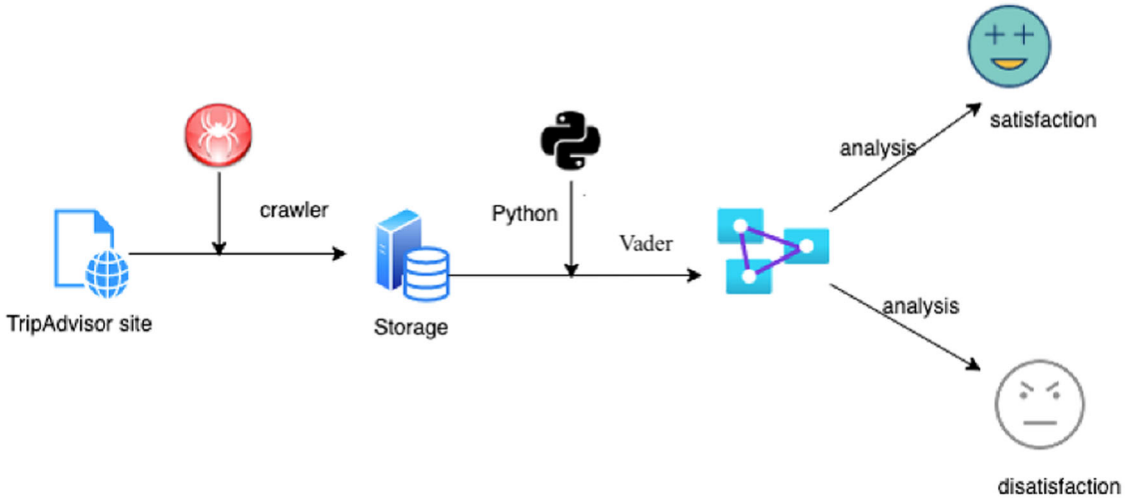


Fig. 1 Measuring customer satisfaction process.

(1) Extract the aspects of Vietnam hotel services that customers need:
Table 1 can be utilized to identify, extract the aspect groups and compute the frequency of words within these groups. These groups comprise the nouns customers mention in their reviews, forming a set of A.

$$Ext(A) = \{\langle a_1, f_1\rangle, \langle a_1, f_1\rangle, \ldots, \langle a_1, f_1\rangle\} \quad (5)$$

In which, a_i are the aspects (services) of Vietnamese hotels and f_j is the number of occurrences of the mentioned aspect in all the reviews.

(2) Percentage of satisfied customers by service aspect

Measure the rate of customer satisfaction according to the service aspects extracted in the previous step according to the following formula:

$$Sas(a_i) = \frac{number\ of\ positive\ reviews(a_i)}{total\ of\ reviews(a_i)} \times 100\% \quad (6)$$

Where :

- Number of positive reviews (a_i): number of positive reviews with aspect a_i
- Total of reviews (a_i): total reviews about aspect a_i

As an illustration, out of 296 reviews that mention the aspect of "room," 259 of them are positive. Therefore, the customer satisfaction rate for the room aspect is calculated to be 87.50% (259/296).

In this research, Python programming language was utilized for data analysis techniques. The study collected and analyzed data from the online booking website with textual data from customer reviews. Using this data, an algorithm model was developed for mining, transforming the textual data into a quantifiable digital format for measurement. The goal was to extract the service aspects that satisfied customers, following specific steps.

- Step 1. Filter positive reviews
- Step 2. Extract the nouns in it
- Step 3. Find the nouns that appear with frequencies > threshold δ
- Step 4. Filter and display the nouns in order of occurrence frequency from high to low. Evaluate the aspects that are most satisfied by customers.

Table 3 List of hotels for data crawling.

No	Name of hotel	Star level
1.	Thang Loi Hotel	4
2.	Hanoi Lotte Hotel	5
3.	Bao Son Hotel	4
4.	Danang Intercontinental Hotel	5
5.	Rex Hotel	5
6.	Star City Nhatrang Hotel	3
7.	Imperial Hue Hotel	3
8.	Melia Hanoi	5
9.	Hanoi Deawoo Hotel	5
10.	Hanoi Hilton Opera Hotel	5
11.	Fortuna Hotel	4
12.	Intercontinental NhaTrang	5

Data collection and data analysis results

Description of data. This study analyzed customer online reviews from TripAdvisor, a trusted travel platform with over 200 million monthly visits and nearly 100 million members worldwide. Travelers from around 80 countries have used the site, generating over 500 million reviews. Hotels and travel companies use TripAdvisor to monitor customer feedback from those who have used their services.

This research only gathered customer reviews from TripAdvisor regarding 12 hotels in Vietnam with 3–5 star ratings. The hotel industry is a specialized service where guests are willing to pay more for high-quality services. Higher-rated hotels (4–5 stars) typically have 1000–2000 reviews as customers in this segment demand exceptional services. In contrast, lower-rated hotels (1–3 stars) usually have fewer reviews (ranging from several dozen to several hundred reviews) and rarely touch 1000 reviews. Hotels rated 1–2 stars infrequently appear on TripAdvisor, and when they do, there are only a few guest reviews, and sometimes, the information is outdated. Also, in Vietnam, the Vietnam National Administration of Tourism manages 4–5 star hotels for quality, while the province's tourism department governs hotels rated 1–3 stars. Table 3 contains the list of hotels used for data collection.

The WebHavy tool collects data and saves it as a .csv file. The collected information includes the name of the reviewer, the date of the review, the content of the review, and the title of the review. Python was used to preprocess the data. The processed data

consists of 20,551 reviews, comprising 2,268,646 words and a vocabulary size of 32,687 words.

From a total of 20,551 reviews collected from 3–5 star hotels in Vietnam. The longest review is 10,335 words, and the shortest is one word. Table 4 lists the data and length of downloaded reviews.

Measuring overall satisfaction of customers with Vietnamese hotel. To analyze sentiments, the Vader library was used to compute positive, negative, neutral, and compound scores (Hutto & Gilbert (2014)). A breakdown of the scores for each evaluation is provided in Table 5.

- Neg (negative): negative score
- Neu (Neutral): score represents neutral
- Pos (Positive): the score represents positivity
- Compound: compound score

Based on the data analysis, dissatisfied customers write shorter reviews, whereas satisfied customers usually write longer sentences. This suggests a positive correlation between the review length and customer satisfaction.

Table 4 Statistics on the length of reviews.

	Length
Count	20,551
Mean	7746.76
Std	3338.97
Min	1.00
Max	10335

	description_lengths	Satisfaction_value
description_lengths	1	–
Satisfaction_value	0.12381021	1

Figure 2 below illustrates the correlation between the length of reviews and customer satisfaction. It also shows the sparseness of reviews on the left. The dense concentration on the right side of the chart means that the number of satisfied customers with Vietnamese hotels accounts for a high percentage.

Based on the compound score, we determine whether customers are satisfied or dissatisfied based on the compound score, and present the statistics in Table 6.

Satisfaction and dissatisfaction rates of customers after measuring are equivalent: satisfied: 94.9%, dissatisfied: 4.75% and 0.35% are not clearly expressed attitudes. The measurement results are shown in Fig. 3.

Measuring the satisfaction of the customer by service aspect. The hotel aspects that make customers satisfied will be found in the positive reviews. As an overview is depicted in Fig. 6, words that occur with high frequency in positive reviews include hotel service aspects such as room, staff, food, service, place, etc. It is easy to see that customers are most satisfied in terms of room, staff, and buffet when they stay in 3–5 star hotels (See Fig. 4).

When extracting the aspects that are most interesting to customers in detail, ten aspects are mentioned with the highest time in the positive reviews.

The chart shown in Fig. 5 displays 10 different aspects that have been mentioned in reviews. Out of all the aspects, the aspect "room" has been mentioned the most, with a total of 15,362 occurrences in reviews. Considering the fact that this aspect has

Table 5 Score of reviews.

Text	Score	Compound
My husband and I traveled here for business and vacation. The Avani was spectacular. Gorgeous views, beautiful spacious rooms and amazing service. We also got to experience the sister hotel Anantara as this time of year is lower occupancy. I felt like a celebrity. The infinity pool by the ocean is incredible. The breakfast was phenomenal. Everything you could imagine from traditional to Western choices. I used the spa twice, facial and pedicure, and had a free yoga and pilates class two mornings. We also took a cooking class which we loved. The executive chef was so kind. Wonderful!!! A local fishing village is only 5 min down the beach. I appreciated the mix of luxury and seeing a more authentic village. This hotel is isolated and quiet. The city is only about a 15 min taxi ride and was also fun to explore. The hotel calls taxis for you and prices are what they say. No dishonesty or haggling. Can't say enough about this place. You will not be disappointed!	{'neg': 0.033, 'neu': 0.77, 'pos': 0.197, 'compound': 0.9885}	0.9885
Nice location and rooms are spacious with sea views, food isn√it that great/overpriced and water was very expensive. After traveling on vietage with amazing service, expectations were high which resort didn√it quite reach, so overall experience just felt that hotel prices weren√it worth it, compared to other resorts we√ive stayed in Vietnam. Staff tried but overall need more training.	{'neg': 0.0, 'neu': 0.888, 'pos': 0.112, 'compound': 0.6055}	0.6055
My family and I had stayed in Avani Quy Nhon in 2019 and thoroughly enjoyed the experience. However, this time not so much. The resort has gotten more rundown and the staff not as well trained. When we arrived, the check in process felt a bit strange and unprofessional. Three staff came to us speaking all at once about multiple things. It was uncomfortable sitting in the lobby that was hot and full of flies. The rooms weren't the cleanest, there was hair in our welcoming dessert, multiple dead lightbulbs in both our rooms and no toilet paper in one of the rooms. The private beach still looked as awesome as I remembered; however, it's no longer as "private" as it once was. Multiple new resorts have sprouted up right beside Avani and was visible from the shoreline. It just felt like any other crowded resort town. During our stay, there was an incident where one of the security guards acted unprofessionally towards us. Management stepped in timely; however, it further soured our experience. With how far this place actually is away from Quy Nhon City and how many new options out there, I don't think it's worth it.	{'neg': 0.09, 'neu': 0.835, 'pos': 0.075, 'compound': −0.3153}	−0.3153

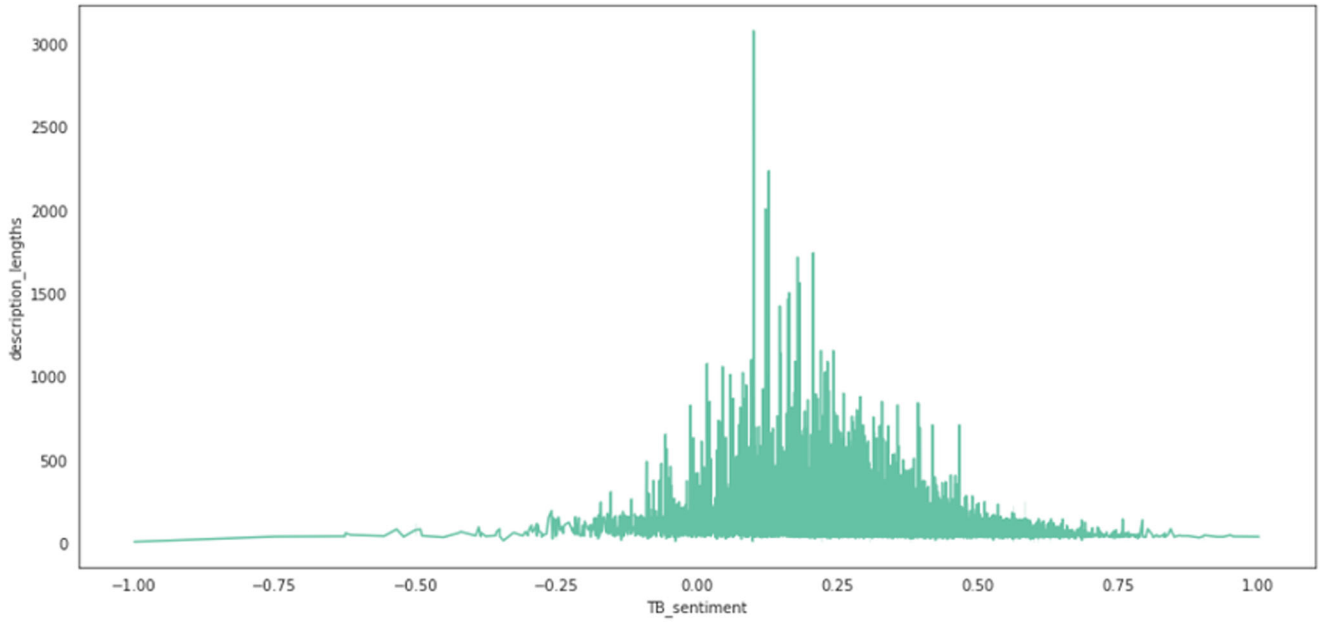


Fig. 2 Length of reviews and customer satisfaction.

Table 6 Classify satisfied or dissatisfied customers.

Category	Number of reviews
Satisfaction	19,504
Dissatisfaction	976
Neutral	71

been mentioned 15,362 times out of 20,551 reviews, it can be concluded that customers argue that the aspect "room" is the most noteworthy when staying in Vietnamese hotels. The next most frequently mentioned aspect is "staff", which has been mentioned 11,937 times by customers.

Figure 5 displays the top 10 aspects that customers find interesting in a hotel. However, it is important to note that these aspects reflect the hotel aspects that interest customers but It does not mean that these aspects make customers satisfied. To measure customer satisfaction for each aspect, we utilize the formula (6).

$$Sas(a_i) = \frac{number\ of\ positive\ reviews(a_i)}{total\ of\ reviews(a_i)} \times 100\%$$

Assume that there are 12,000 reviews. Consider the aspect "room": there are 7224 positive reviews mentioned it; 1203 positive reviews did not mention the aspect "room"; the aspect "room" is also mentioned many times in 20 positive reviews. In this case, it seen that is one. The number of positive reviews is $= 7224 + 1203 + 20 = 8447$. Sas(room) $= 70.39\%$.

In Fig. 6, we can see the percentage of customers interested in various aspects of the hotel. We can identify the service aspects that attract customers by analyzing them closely. This helps us understand which aspects are satisfying for customers and which are not. Hotels can work on improving and upgrading their service quality accordingly.

We analyzed more than 20,000 reviews and found that customers show the most interest in the room service when they stay at hotels in Vietnam. The staff aspect also receives high ratings from customers.

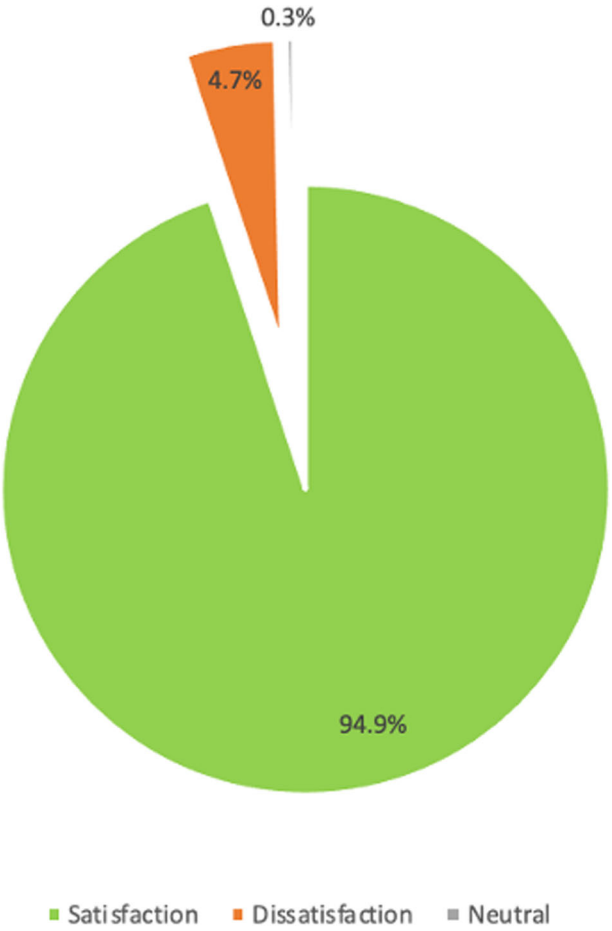


Fig. 3 Satisfaction and dissatisfaction rates with 3–5 stars hotels in Vietnam.

Measuring customer satisfaction by lexical rule. In order to delve deeper into the satisfaction levels of customers with regard to different aspects of the service, this study utilizes polarity words in combination with those aspects. Polarity words refer to



Fig. 4 WordCloud of the most frequent aspect that the customer mentions.

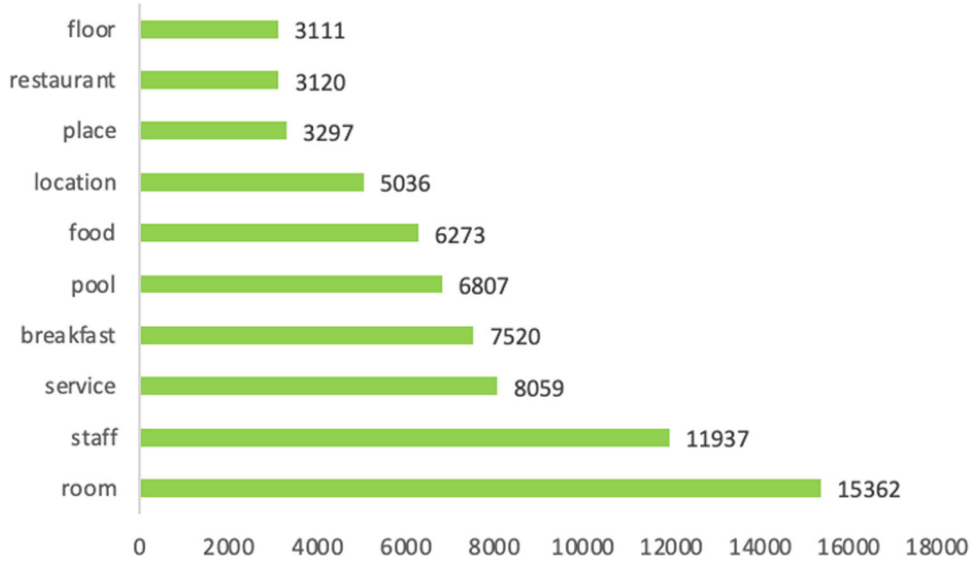


Fig. 5 Extracting the highest frequent aspects.

adjectives and adverbs that convey the emotional attitude of customers towards each aspect. Typically, these adjectives and adverbs are placed either to the left or right of the relevant aspect to indicate the level of satisfaction associated with it.

Table 7 lists polarity words that indicate customer satisfaction in reviews. These words, known as sentiment words, often include positive adjectives like good, great, nice, excellent, and friendly. These adjectives are frequently found in positive reviews.

We analyze the correlation between aspect words and polarity words by examining their frequency of occurrence together. This data reveals a noticeable difference in customer satisfaction levels across different service aspects. Figure 7 provides an example of how aspect and polarity words can be combined.

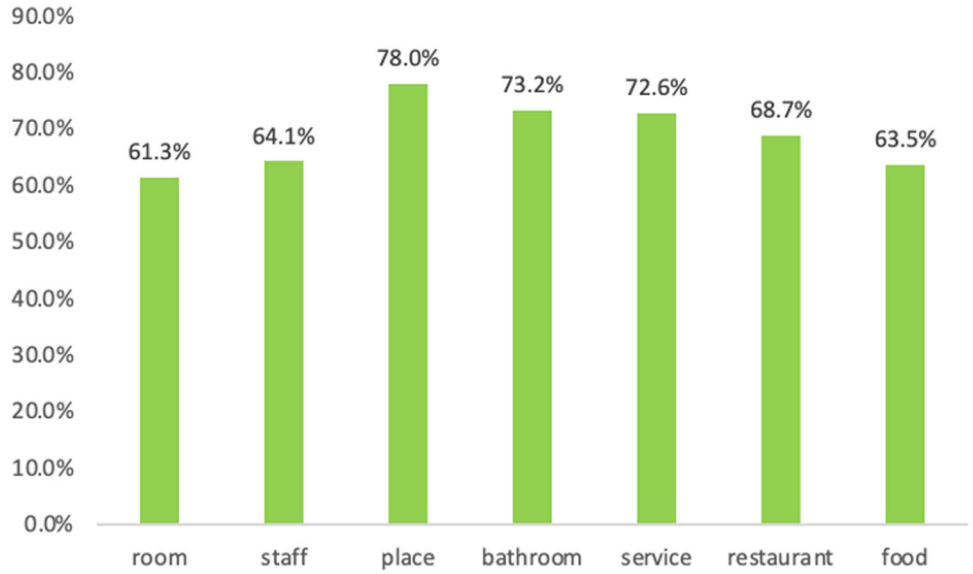


Fig. 6 The list of aspects with the ratio that are satisfied by customers.

Table 7 List of polarity words in positive reviews.

No	Words	Frequency	#Reviews that occur
1.	Good	0.374	3945
2.	Great	0.318	3938
3.	Nice	0.213	2717
4.	Excellent	0.162	2181
5.	Friendly	0.161	2495
6.	Clean	0.148	2079
7.	Helpful	0.145	2010
8.	Best	0.115	1739
9.	Beautiful	0.100	1831
10.	Lovely	0.083	1165

When calculating the combination of words, we follow these steps:

- Separating of sentences in all of the positive reviews.
- Filter sentences with a predefined aspect such as room, staff, place, and services. For example, separate all sentences containing the aspect "room".
- Locate the aspect in the sentence. Example sentences: "And I am very happy with the nice room and beautiful place." then the word "room" is in position 9 in the sentence containing it.
- Cut the string from the position of aspect to the left and right aspects to 4 words. For example, the string on the left is "with the nice room", and the string on the right is "room and beautiful place".
- Search in the left string and right string, and the left string first. Count frequency when meeting the first polarity word and stop when meeting another aspect. Check the right string when there is no polarity word in the left string.
- List all the polarity words associated with each aspect and their pairing frequency.

Table 8 below lists some aspects combining polarity words and their frequency.

Identifying the dissatisfaction aspects of hotel service. Negative reviews of hotels often come from dissatisfied customers. Figure 8 below displays a word cloud of the common aspects mentioned in these negative reviews, which include restaurant, resort, pool, room, services, and price. Negative adjectives like understaffed, and badly are frequently used.

To measure customer dissatisfaction with each aspect of hotel service, we use formula (6)

$$Sas(a_i) = \frac{number\ of\ negative\ reviews(a_i)}{total\ of\ reviews(a_i)} \times 100\%$$

Where number of negative reviews (a_i) is the number of positive reviews containing aspect a_i out of the total number of reviews containing aspect a_i. The result is shown as the Fig. 9. below

Based on the Fig. 9 presented, it is evident that the most common complaint among customers is related to the "room" aspect, followed by the "staff" aspect which also accounts for 30.3%. Additionally, the "price" aspect is a significant source of customer dissatisfaction as it is among the highest levels of complaints.

Discusion and conclusion

Understanding customer interests through their reviews has become an important aspect of business (Akhtar et al., 2017; Alaei et al., 2019; Breda et al., 2020). Numerous studies have explored various approaches to this issue (Alrawadieh & Law, 2019; Barnes et al., 2020). Natural language processing techniques provide different ways of analyzing customer reviews (Chalupa & Petricek, 2022; Chen et al., 2019; Dominy et al., 2022; Dong et al., 2014; Lee et al., 2019; Padma & Ahn, 2020). Our study offers clearer definitions, formulas, and processes than previous studies (Godnov & Redek, 2018). We also use inferential statistics and language rules, such as adverbs, adjectives, and nouns in their respective positions in sentences, as well as combining positive words. By focusing on rule-based sentences, we discovered issues hidden by semantic ambiguity in reviews, which are difficult to detect with conventional mathematical models.

Some important discoveries made in this research are:

- There is a positive relationship between sentence length and customer satisfaction. Satisfied customers tend to write lengthy reviews.
- When measuring satisfaction at the overall level, the customer satisfaction rate reached 94.9%, and dissatisfaction:

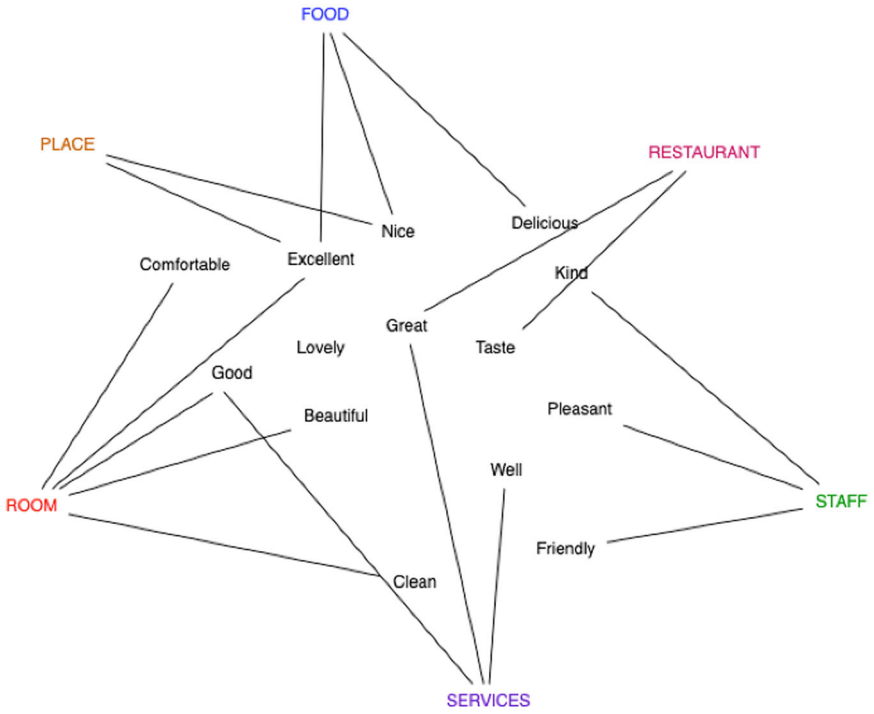


Fig. 7 The pairing of aspects and polarity words.

Table 8 Frequency of aspect words combined with polarity words.

Aspect	Polarity	# Occurrance
Room	Beautiful	223
	Nice	184
	Well	256
	Large	323
	Comfortable	371
	Great	300
	Excellent	197
	Clean	749
	Wonderful	56
	Good	428
Staff	Friendly	613
	Helpfull	902
	Incredibly	49
	Attentive	210
	Nice	288
	Enthusiast	13
	Kind	94
	Absolutely	27
	Pleasant	91
	Lovely	143
	Good	315
Food	Great	398
	Excellent	290
	Delicious	194
	Good	584
	Lovely	62
	Taste	168
	Fantastic	91
	Amazing	125
Restaurant	Great	228
	Excellent	183
	Good	192
	Nice	264
	Well	106
	Amazing	126
	Full day	29
	Diversity	132

4.75%. However, when measuring the aspect level, the aspect "place" has the highest satisfaction but only 78%. This shows that it is necessary to analyze the aspect level to show which aspect is most satisfied by customers clearly,

- The aspect "room" is the most interesting to customers, but only 61% of satisfied customers. Therefore, it is necessary to consider upgrading the quality of the "room" service.
- Detect pairs of polarity associated with aspect words with frequency. From there, it is possible to analyze more deeply to measure customer satisfaction at different levels (for example, on a scale from 1–5, we will continue to research in the future).
- The aspect "price" appears in the highest levels of customer complaints.

Vietnam is a developing country located in Southeast Asia. Over the past few years, Vietnam has become the second most popular destination in the region for international tourists. To further enhance the hospitality industry in Vietnam, hotel managers should increase their online presence and strengthen their global governance, especially for 1–3 star hotels. By taking advantage of affordable and effective opportunities, such as updating brand information and communicating through online booking platforms, hotels can better reach their customers. Moreover, in today's highly connected social environment, social listening is crucial for uncovering customer opinions across various interactive channels. Social listening is a market research model that helps hotels to better understand their customers' psychology and emotional views, assess customer satisfaction with hotel services, and manage media crisis situations that may affect the hotel's reputation. The results of the study show that there are many hidden meanings in positive reviews. Discovering information from customer reviews helps manage hotel quality better. However, when the data is so massive, using textual data analysis techniques becomes meaningful for hotels with thousands of reviews on social platforms.

In this paper, we present a method for analyzing customer reviews on online booking websites. This approach helps to comprehend the sentiments and viewpoints of customers who have utilized Vietnamese hotel services. Many hotels globally are adopting this innovative approach to improve customer



Fig. 8 Wordcloud of negative words.

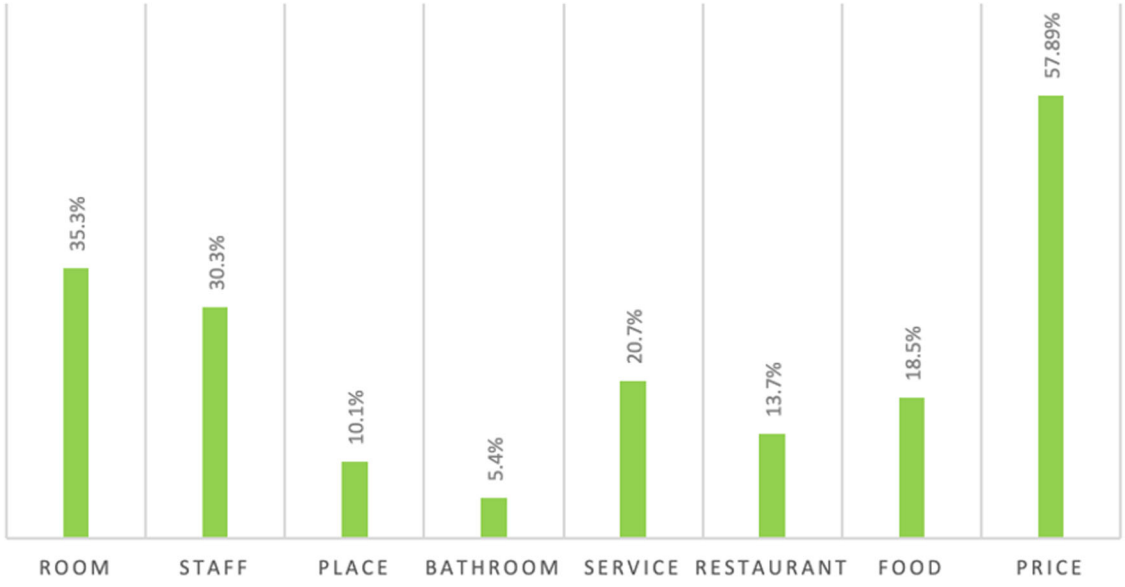


Fig. 9 The ratio of unsatisfied aspects.

satisfaction. Data analysis is a new direction that many hotels worldwide are approaching. Applying this approach shows that customer satisfaction for 3–5 star hotels in Vietnam at the overall level is higher than the aspect level. Customers' satisfaction is in some aspects such as room, staff, restaurant, and food. However, the limitation of this study is that it needs to be an in-depth analysis of the unsatisfied aspects of guests. These limitations will be our future work. So in the future, we will use linguistic rules at a deeper level and analyze polarity words on the negative side to see what guests complain about the hotel. Our future studies will focus on language rules and polarity words more deeply. It expects to solve problems of language ambiguity to bring about more accuracy when analyzing online customer reviews.

Data availability

The datasets generated during and analyzed during the current study are available from the corresponding author upon reasonable request.

Received: 25 November 2022; Accepted: 7 September 2023;
Published online: 27 September 2023

References

Akhtar N, Zubair N, Kumar A, Ahmad T (2017) Aspect based sentiment oriented summarization of hotel reviews. Proc Comput Sci 115:563–571. https://doi.org/10.1016/j.procs.2017.09.115

Alaei AR, Becken S, Stantic B (2019) Sentiment analysis in tourism: capitalizing on big data. J Travel Res 58(2):175–191. https://doi.org/10.1177/0047287517747753

Alrawadieh Z, Law R (2019) Determinants of hotel guests' satisfaction from the perspective of online hotel reviewers. Int J Cult Tour Hosp Res 13(1):84–97. https://doi.org/10.1108/ijcthr-08-2018-0104

Barnes SJ, Mattsson J, Sørensen F, Jensen JF (2020) Measuring employee-tourist encounter experience value: a big data analytics approach. Expert Syst Appl 154(113450):113450. https://doi.org/10.1016/j.eswa.2020.113450

Blomberg-Nygard A, Anderson CK (2016) United nations world tourism organization study on online guest reviews and hotel classification systems: An integrated approach. Serv Sci 8(2):139–151. https://doi.org/10.1287/serv.2016.0139

Chalupa, S, & Petricek, M (2022). Understanding customer's online booking intentions using hotel big data analysis. J Vacat Mark https://doi.org/10.1177/13567667221122107

Chen M-C, Hsiao Y-H, Chang K-C, Lin M-K (2019) Applying big data analytics to support Kansei engineering for hotel service development. Data Technol Appl 53(1):33–57. https://doi.org/10.1108/dta-05-2018-0048

Chowdhary, KR (2020). Natural Language Processing. In Fundamentals of Artificial Intelligence (pp. 603–649). Springer India

Li H, Liu Y, Tan C-W, Hu F (2020) Comprehending customer satisfaction with hotels: Data analysis of consumer-generated reviews. Int J Contemp Hosp Manag 32(5):1713–1735. https://doi.org/10.1108/ijchm-06-2019-0581

Dominy CL, Arvind V, Tang JE, Bellaire CP, Pasik SD, Kim JS, Cho SK (2022) Scoliosis surgery in social media: a natural language processing approach to analyzing the online patient perspective. Spine Deformity 10(2):239–246. https://doi.org/10.1007/s43390-021-00433-0

Dong J, Li H, & Zhang X (2014). Classification of customer satisfaction attributes: An application of online hotel review analysis. In IFIP Advances in Information and Communication Technology (pp. 238–250). Springer Berlin Heidelberg

Breda, Z, Costa, R, Dinis, G, & Martins, AA (2020). EWOW of guests regarding their hotel experience: Sentiment analysis of TripAdvisor reviews. In Handbook of Research on Social Media Applications for the Tourism and Hospitality Sector (pp. 295–308). IGI Global

Godnov U, Redek T (2018) Good food, clean rooms and friendly staff: Implications of user-generated content for Slovenian skiing, sea and spa hotels' management. Manag J Contemp Manag Issues 23(1):29–57. https://doi.org/10.30924/mjcmi/2018.23.1.29

Gómez-Suárez, M, & Veloso, M (2022). Designing Facebook publications focused on hotel customer experience: How to improve brand attitude and booking intention. In Brand, Label, and Product Intelligence (pp. 247–258). Springer International Publishing

Hu N, Zhang T, Gao B, Bose I (2019) What do hotel customers complain about? Text analysis using structural topic model. Tour Manag 72:417–426. https://doi.org/10.1016/j.tourman.2019.01.002

Hutto, CJ & Gilbert, EE (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. Eighth International Conference on Weblogs and Social Media (ICWSM-14)

Lee M, Lee S (ally), Koh Y(2019) Multisensory experience for enhancing hotel guest experience: Empirical evidence from big data analytics Int J Contemp Hosp Manag 31(11):4313–4337. https://doi.org/10.1108/ijchm-03-2018-0263

Leal F, Malheiro B, Burguillo JC (2019) Analysis and prediction of hotel ratings from crowdsourced data. Wiley Interdiscip Rev Data Min Knowl Discov 9(2):e1296. https://doi.org/10.1002/widm.1296

Li Q, Li S, Zhang S, Hu J, Hu J (2019) A review of text corpus-based tourism big data mining. Appl Sci 9(16):3300. https://doi.org/10.3390/app9163300

Liu Y, Teichert T, Rossi M, Li H, Hu F (2017) Big data for big insights: investigating language-specific drivers of hotel satisfaction with 412,784 user-generated reviews. Tour Manag 59:554–563. https://doi.org/10.1016/j.tourman.2016.08.012

Luturlean, BS, & Anggadwita, G (2016). A framework for conceptualizing customer experiences management in the hotel industry. *Proceedings of the 3rd International Seminar and Conference on Learning Organization (Isclo-15)*

Mohamed ESA (2021) The impact of customer experience and relationship quality on corporate reputation in the hotel sector. Int J Cust Relationsh Mark Manag 12(2):53–79. https://doi.org/10.4018/ijcrmm.2021040104

Padma P, Ahn J (2020) Guest satisfaction & dissatisfaction in luxury hotels: an application of big data. Int J Hosp Manag 84(102318):102318. https://doi.org/10.1016/j.ijhm.2019.102318

Paulose D, Shakeel A (2022) Perceived experience, perceived value and customer satisfaction as antecedents to loyalty among hotel guests. J Qual Assur Hosp Tour 23(2):447–481. https://doi.org/10.1080/1528008x.2021.1884930

Rahimian S, ShamiZanjani M, Manian A, Esfidani MR (2021) A framework of customer experience management for hotel industry. Int J Contemp Hosp Manag 33(5):1413–1436. https://doi.org/10.1108/ijchm-06-2020-0522

Sampetua Hariandja E, Vincent F (2022) Linking customer experience, satisfaction, and loyalty to brand power and performance in international hotels. Innov Mark 18(3):59–71. https://doi.org/10.21511/im.18(3).2022.06

Thu HNT, Minh TT, Binh GN, Ngoc VH (2020) Determining criteria for evaluating quality of Vietnamese hotel through guest s online reviews. Int J Bus Inf Syst 1(1):1. https://doi.org/10.1504/ijbis.2020.10036288

Vo NT, Hung VV, Tuckova Z, Pham NT, Nguyen LH (2022) Guest online review: an extraordinary focus on hotel users' satisfaction, engagement, and loyalty. J Quali Assurance Hosp Tour 23(4):913–944

Zhao Y, Xu X, Wang M (2019) Predicting overall customer satisfaction: Big data evidence from hotel online textual reviews. Int J Hosp Manag 76:111–121. https://doi.org/10.1016/j.ijhm.2018.03.017

Author contributions

H.N.T.T. built a research model and conducted the simulations; T.N.X. raised the research question and collected data; T.N.X. and H.N.T.T. analyzed the data and they wrote the paper.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Ha Thi Thu Nguyen or Trung Xuan Nguyen.

Reprints and permission information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this license, visit http://creativecommons.org/licenses/by/4.0/.

© The Author(s) 2023